INFORMATION SECURITY PROJECT REPORT

Assignment

Security Analysis and Implementation in Smart Home IoT System

Bachelors of Technology

In

Electronics & Communication Engineering

By

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1. Domain Selection: Internet of Things (IoT)

The Internet of Things (IoT) refers to a network of interconnected physical devices that collect, exchange, and process data over the internet. These devices include smart home appliances, wearable devices, industrial sensors, and healthcare monitoring systems.

In this project, we focus on a **Smart Home IoT System**, which includes devices such as smart locks, temperature sensors, and lighting systems. These devices communicate with users through mobile or web interfaces and often rely on cloud-based services.

IoT systems are highly vulnerable to security threats due to limited computational resources, lack of proper security mechanisms, and continuous connectivity. Therefore, they provide an ideal domain for studying information security.

2. Security Problems Identified

2.1 Unauthorized Access

Many IoT devices use weak or default credentials, making them susceptible to unauthorized access. Attackers can gain control over critical devices like smart locks and cameras.

2.2 Data Leakage and Privacy Issues

IoT devices collect sensitive user data such as daily routines and environmental conditions. If data is transmitted without encryption, it can be intercepted and misused.

2.3 Weak Communication Security

Communication between IoT devices often occurs over insecure protocols like HTTP. This exposes the system to Man-in-the-Middle (MITM) attacks and data manipulation.

2.4 Malware and Botnet Attacks

Due to limited security capabilities, IoT devices can be infected with malware and used as part of botnets to launch large-scale attacks.

2.5 Lack of Firmware Updates

Many IoT devices do not receive regular updates, leaving known vulnerabilities unpatched and exploitable.

3. Proposed Security Mechanisms

3.1 Strong Authentication

A secure login system using hashed passwords (SHA-256) is implemented to prevent unauthorized access.

3.2 Data Encryption (AES)

Sensitive data is encrypted using AES encryption to ensure confidentiality during transmission.

3.3 Secure Communication with Integrity Check

Hashing techniques (SHA-256) are used to verify data integrity and prevent tampering.

3.4 Access Control

Role-based access control is implemented to restrict critical operations such as unlocking doors to authorized users only.

3.5 Secure Update Validation

Firmware updates are validated using hashing techniques before installation to ensure authenticity.

4. System Architecture Diagram

```
+-------------+        +-------------------+        +-------------------+
|  Mobile App | -----> |  IoT Gateway/App  | -----> |   Smart Devices   |
| (User CLI)  | HTTPS  | (Encryption/Hash) |  LAN   | (Lock, Sensor)    |
+-------------+        +-------------------+        +-------------------+
        ^                           |
        |                           v
        |                  +-------------------+
        |                  |   Cloud/Server    |
        |                  | (Storage/Logs)    |
        |                  +-------------------+
```

Figure 1: IoT System Architecture

Description: User sends commands via CLI → encrypted and hashed at gateway → delivered to devices → optional logging to cloud.

5. Data Flow Diagram

```
User Input
   |
   v
[Login/Auth] --(SHA-256)--> [Verify Credentials]
   |
   v
[Create Command]
   |
   v
[Encrypt Command (AES/Fernet)]
   |
   v
[Generate Hash (SHA-256)]
   |
   v
====== Transmission ======
   |
   v
[Receive Data]
   |
   v
[Verify Hash]
   |        \
   |         \---> If mismatch → Reject (Tampered)
   v
[Decrypt Command]
   |
   v
[Access Control Check]
   |        \
   |         \---> If unauthorized → Deny
   v
[Execute Command]
```

Figure 2: Data Flow Diagram

6. Implementation

6.1 Tools and Technologies Used

- Programming Language: Python
- Libraries: hashlib, cryptography (Fernet for AES encryption)

6.2 System Features

- User authentication with hashed passwords
- Encryption and decryption of commands
- Data integrity verification using hashing
- Role-based command execution

6.3 Working of the System

1. The user logs into the system using a username and password.
2. The entered password is hashed and compared with stored hashed values.
3. Upon successful authentication, the user can send commands to IoT devices.
4. The command is encrypted before transmission.
5. A hash of the command is generated to ensure integrity.
6. On the receiver side, the command is decrypted and the hash is verified.
7. If the integrity check passes, the command is executed based on user role.

7. Flowchart of Implementation

```
Start
  |
  v
Login?
  |----No----> Exit
  |
 Yes
  |
  v
Menu (Send Command / Exit)
  |
  v
Enter Command
  |
  v
Encrypt + Hash
  |
  v
Send → Receive
  |
  v
Hash Valid?
  |----No----> Reject
  |
 Yes
  |
  v
Decrypt
  |
  v
Authorized?
  |----No----> Deny
  |
 Yes
  |
  v
Execute
  |
  v
Repeat / Exit
```

Figure 3: System Flowchart

8. Results and Analysis

The implemented system successfully demonstrates key information security principles:

- **Confidentiality:** Achieved through AES encryption of data.
- **Integrity:** Ensured using SHA-256 hashing.
- **Authentication:** Implemented via secure login system.
- **Authorization:** Enforced using role-based access control.

The system prevents unauthorized access, detects tampering, and ensures secure communication between components.

9. Conclusion

This project demonstrates how basic but effective security mechanisms can be applied to IoT systems to mitigate common threats. By implementing authentication, encryption, and integrity checks, the Smart Home IoT system becomes significantly more secure.

The project highlights the importance of integrating security at the design level in IoT applications and provides a foundation for further enhancements such as intrusion detection systems and secure firmware updates.

10. Future Scope

- Implementation of Intrusion Detection System (IDS)
- Integration with real IoT hardware devices
- Use of advanced encryption techniques (RSA, ECC)
- Secure cloud integration

11. References

- Cryptography Documentation (Python)
- IoT Security Research Papers
- Online educational resources on Information Security

12. GitHub Link

GitHub Repository Link:

https://github.com/ YajatModi09/secure-smart-home-iot